Compact Lightweight Aerial Sensor System (CLASSy)

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Abstract

In the wake of increasingly intense wildfires, innovative solutions are imperative to enhance wildfire mitigation strategies. Current technological integrations have hit a communicative limit. Between limited flight time, computational expenses as well as financial expenses, there is a hole in the market for an effective, low-tech, and disposable solution. The Compact Lightweight Aerial Sensor System (CLASSy) is designed to revolutionize active disaster operations through comprehensive decision support. CLASSy consists of a lightweight launch mechanism and a flight body equipped with a sensor package and parachute. The assembly integrates sensor networks with data analytics to provide real-time, high-resolution information to incident commanders, directly facilitating decision-making and resource allocation. Infrared imagery and temperature differentials are processed and analyzed throughout flight, offering valuable insights into fire behavior, hotspot detection, and fire spread trajectories. CLASSy is intended to meet a variety of natural disaster mitigation needs through its variable launch height and disposability. CLASSy's goal is to assist wildfire fighting without taking up any human or material resources. As a result, CLASSy is as lightweight as possible, easily expendable, inexpensive to manufacture, and only requires one operator for effective use. CLASSy's integrated sensor suite, real-time analytics, and closed loop active communications empower firefighting teams to proactively address wildfire challenges. As the frequency of wildfires continues to rise, technological innovations like CLASSy are crucial to effective wildfire management systems.

Nomenclature

Abbreviations

- CAD – Computer Aided Design
- PLA – Polylactic Acid
- FOV – Field of View
- EPA – Environmental Protection Agency
- ConOps – Concept of Operations
- PWM – Pulse Width Modulation
- BLE – Bluetooth Low Energy
- W – Wireless
- LCD – Liquid-crystal Display
- GPIO – General Purpose Input/Output
- CG – Center of Gravity
- CP – Center of Pressure
- ID – Inner Diameter
- OD – Outer Diameter

Symbols

- g = gravity
- m = mass
- h = height
- k = elastic constant
- KE = kinetic energy
- PE = potential energy
- t = time
- V = velocity
- d = diameter
- W = weight
- C_d = coefficient of drag
- ρ = air density
- x = displacement

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I. Introduction

A. Background

With the changing climate, wildfires have become more detrimental to the environment, burning increasingly more acreage each year since the 1980s [1]. Although wildfires serve a vital role in many different ecosystems, allowing important nutrients and other organic material the opportunity to re-enter the soil, they have become extraordinarily more violent and damaging. Uncontrolled wildfires are responsible for removing important root systems, leading to contaminated water systems due to run off. Additionally, they degrade the overall air quality of anything downwind of the affected area, especially in drought-prone regions [2]. Current wildfire fighting operations rely on delayed data, resulting in inefficiencies in containing and managing wildfires on the front line. Last-mile connectivity poses the greatest obstacle in relaying real-time updates to firefighters. To combat this pressing issue, a deployable Compact Lightweight Aerial Sensor System (CLASSy) has been designed for first responders to use on the front line.

B. Project Overview

CLASSy is a low-tech, cost-effective disaster mitigation tool that provides decision support for firefighters based on fire perimeter detection from an aerial point of view. CLASSy is designed to be carried and operated by a singular boots-on-the-ground firefighter on the frontline. CLASSy uses a reusable elastic launch system with expendable launch units that are each equipped with an IR sensor package. This sensor package collects IR images to visualize perimeters and velocities of wildfire fronts as the device slowly descends via parachute following apogee. The collected data is then transmitted back to ground via Bluetooth to a data display which provides real-time updates on fire spread velocity.

II. Specifications and Constraints

A. Overall Requirements and Design Constraints

CLASSy's system requirements are primarily driven by stakeholder needs. The stakeholder is identified as any natural disaster relief organization (e.g., CAL FIRE). Current relief efforts rely on expensive, piloted drones and other higher altitude aircraft to provide decision support for ground crews. Consultations with stakeholders established expectations about what could be improved for higher efficiency relief operations. These desired improvements in operations established an operations concept with support strategies visualized in Figure 1.

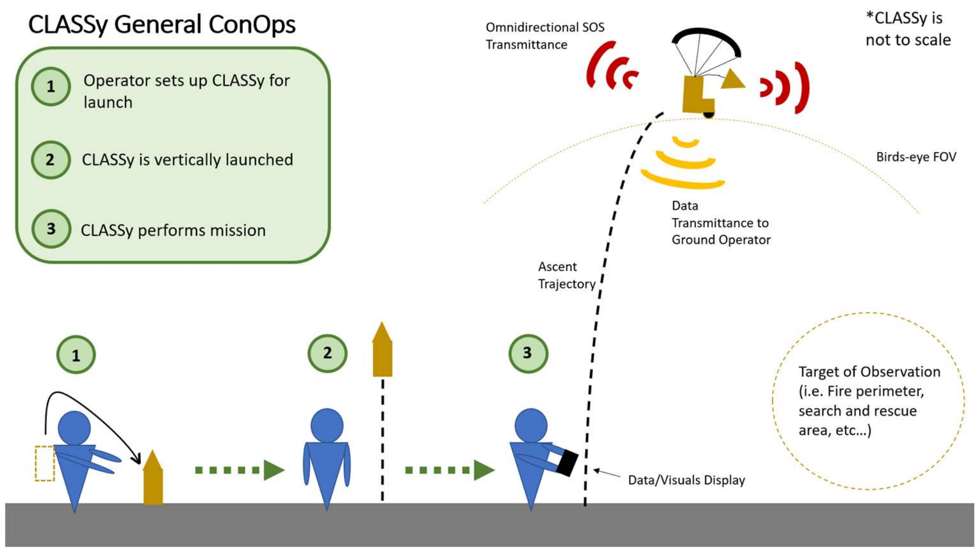


Figure 1 Concept of operations for disaster relief solution.

A set of use cases for CLASSy drives the creation of an alternative mission concept of operations which is found in Appendix A. From these alternative missions, the worst-case scenario is deemed to be wildfire fighting operations which is visualized in Figure 2. This worst-case scenario helps drive the design requirements for CLASSy.

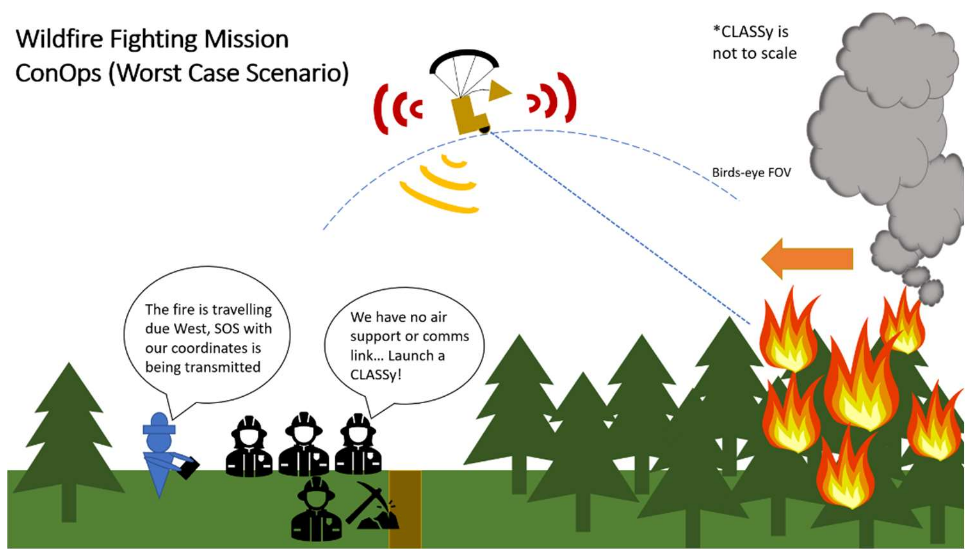


Figure 2 Mission concept of operations for worst-case scenario.

The concept of operations is partitioned into three stages of execution. First, the ground operator sets up the launch mechanism of CLASSy with the body frame loaded into it. Next, the body is vertically launched via release of the launch system by the ground operator. Last, CLASSy performs its mission by sending data back to the ground operator through means of a digital display.

From this operations concept, stakeholder expectations are further defined through requirement statements. CLASSy's design is driven by eight system requirements which are all centered around lower costs of manufacturing, ease of use, and passivity of flight controls. These driving requirements and their justifications are in Table 1.

Table 1 CLASSy driving requirements.

Req #	*The device shall…*	**Justification**
R1	Be launched to a max altitude of approximately 200 ft AGL.	Higher altitudes ensure that downward visuals cover enough area to be useful for ground crew.
R2	Have a minimum flight time of 1 minute.	There needs to be enough time in air to provide viable sensor data and establish a communications link.
R3	Weigh less than 5 pounds.	Firefighters do not have the capacity to add extensive amounts of weight to their already heavy packs.
R4	Have a maximum upfront cost of $200 for one launch system with 6 units.	Needs to cost less than current competition which starts at $15,000. Additionally needs to be inexpensive enough to be fully expendable.
R5	Be compact enough for 1 trained ground operator to carry.	Needs to be ergonomic and the size of the package not cumbersome to the operator.
R6	Take 60 minutes of training time to learn how to fully operate.	Needs less training to operate than current competition (e.g. drones).
R7	Be made of materials that are within regulatory (EPA) acceptable limits of environmental toxicity damage.	The design should not cause long term environmental damage by leaching toxins into the environment.
R8	Be multipurpose.	It is advantageous to design a product that can be deployed in all disaster response scenarios.

B. Sub-System Requirements and Design Constraints

To further clarify CLASSy's design constraints, child requirements are derived for each sub-system. Each child requirement is directly sourced from the overall driving requirements. The child requirements for each sub-system with their associated driving requirements and justifications are found in Tables 2-5.

Conversations with experts in the wildfire fighting industry indicated there are several requirements that a useful sensor package needs to meet, as listed in Table 2. Consistent sensor measurements are necessary to detect the speed and direction of a moving fire in a volatile environment. Data taken from the sensors should be processed and displayed in a format that is useful in active decision support. In turn, aerial imaging techniques are employed to recognize vital firefighting landmarks, such as rivers, roads, and ridges. A beneficial interface for active decision support should convey fire trajectory as well as directions to inform operations. The collected information should be transmitted in real time through a reliable wireless communications network between the sensor package and a device on the ground.

Table 2 Electrical system requirements.

Req #	*The electrical system shall…*	Parent Req #	Justification
RE1	Maintain an active communications link through at least 25% of the flight.	R6	Viable data acquisition would occur for approximately the first 25% of descent.
RE2	Be consistent in reading accuracy within 10% of actual.	R6	For data to be relied upon and multipurpose, reading abilities must be consistent in occurrence and resolution.
RE3	Weigh less than 1lb.	R1	Device requires a minimum airtime to complete mission, dependent on system weight.

The parachute system must provide the sensor system enough time and stability to collect and transfer the desired data back to ground operators. Therefore, descent speed, air stability, and parachute size are considered the most pertinent design constraints for the parachute. These constraints are specified into the parachute system requirements which are shown in Table 3.

Table 3 Parachute system requirements.

Req #	*The parachute system shall…*	Parent Req #	Justification
RP1	Provide a descent rate of no less than 15 ft/sec.	R2	There must be enough airtime for the sensor unit to collect and transmit data.
RP2	Be no greater than 13 in^3 when collapsed.	R3	Must be small enough to passively fall out of the body tube at apogee.
RP3	Remain aero mechanically stable in wind gusts up to 40 mph.	R8	Wind gusts around wildfires can frequently get up to 40 mph and the parachute must not fail in these gusts.

To accommodate the sensor package, a reliable and sturdy body is needed. The flight body must be large enough to contain the sensors and recovery system without sacrificing aerodynamics. Similarly, the body must weigh less than one pound, carry the payload to at a specified height, and be disposable. The requirements are further specified in Table 4.

Table 4 Body frame requirements

Req #	*The body frame shall…*	Parent Req #	Justification
RS1	Prevent the destruction of essential components during operation.	R1	Device structure should enable operations in all mission environments.
RS2	Weigh no greater than 1 lb.	R3	It should weigh enough to be structurally sound but not hinder the aerodynamic performance.
RS3	Cost no more than $75 in materials to construct.	R4	It should not be difficult to find/order materials for both NASA or Cal Fire and ideally be environmentally friendly.

As seen in Table 5, the CLASSy system requires a rugged and reliable launching mechanism that enables the rapid deployment of aerial sensor packages during time-sensitive disaster relief scenarios. The mechanism needs to be lightweight to allow for one adult to easily transport it on foot prior to deployment. The system should be simple enough for an individual to deploy in less than five minutes and should require minimum training to operate. Safety is also a paramount consideration during the design process.

Table 5 Launch system requirements.

Req #	*The launch system shall…*	**Parent Req #**	**Justification**
RL1	Be operable by 1 adult.	R5	Firefighters need to be able to continue their work at the same pace and not be slowed down to set up and launch the system.
RL2	Launch to the target altitude.	R1	The sensor needs to reach an adequate height to intake useful data for the ground crew.
RL3	Weigh under 5 lbs.	R3	Firefighters already carry heavy loads and do not have the weight capacity to add heavy objects.
RL4	Have a max dimension of 2 feet.	R5	The mechanism needs to be easily carried by the operator, preferably on top of the pack they already carry.

C. Product Breakdown Structure

The overall and sub-system driving requirements enforce the product breakdown structure. The product breakdown structure visualizes the breakdown of CLASSy into its sub-systems and their constituents. CLASSy is a multi-faceted design comprised of a sensor unit, body frame, launch system, and descent system. The sensor unit contains a microcontroller, equipped with various sensors, and a battery pack for power. The body frame houses the sensor unit and additionally houses the parachute descent system. The launch system is a fully mechanical, elastic, cock-lock-and-release mechanism that launches the body assembly to a variety of desired altitudes. The relationship of CLASSy and its sub-systems is visualized in Figure 6.

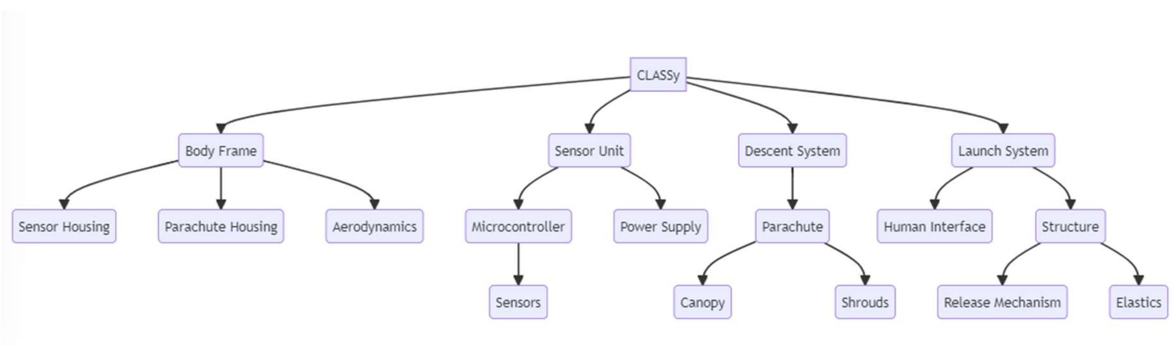


Figure 3 Product breakdown structure visualizing CLASSy's sub-systems.

III. Design Assembly

A series of trade studies is performed to select a method of launching and keeping the sensor unit in the air to satisfy CLASSy's design requirements. The trade studies rank system requirements on priority and separate wants versus needs. The initial trade study explores the use of a balloon, kite, drone, auto-gyro, parachute, and glider to keep the sensor unit aloft. The first two alternatives are self-propelled while the latter four require external power to launch the sensor package. This trade study narrows down potential methods of suspending CLASSy in the air to a balloon, kite, parachute, and auto-gyro. This trade study can be found in greater detail in Appendix B. The results of this trade study support the use of a combined external launch and parachute system to support CLASSy's mission.

A. Sensor Package

1. Sensors

Sensor selection is largely motivated by availability and resolution. To keep the sensor package as lightweight as possible, the team is primarily interested in selecting multifaceted sensors. In creating a device capable of flight, an altimeter or pressure sensor is necessary. Similarly, to assist in the fight against wildfires, a variety of temperature sensitive devices are tested. Infrared sensors, cameras, and temperature sensors are tested for resolution and potential employment. The Sparkfun AMG8833 IR Thermal Camera Breakout, SparkFun MLX90640 IR Array Breakout, Adafruit BMP390 Precision Barometric Pressure and Altimeter, and the Adafruit MCP9808 High-Accuracy Temperature Sensor are each considered. The sensors are assessed through cost effectiveness—what information could the sensor contribute to the system, and is that worth its cost? To prioritize the disposability of the system, the entire sensor package needs to be as low cost as possible without sacrificing the mission of the system. Ultimately, the AMG8833 IR Camera breakout and the BMP390 are integrated to the final design.

An accessible microcontroller is considered to accommodate the selected sensors. Due to its accessibility, inexpensive cost, and adaptability, the Raspberry Pi Pico W is selected. The microcontroller is programmable using CircuitPython and MicroPython, two Python derivative programming languages. Similarly, the Pi Pico W has 26 multifunction GPIO pins with 2 I2C channels and 16 PWM channels. Given the sensor foundation of the system, access to these features is vital. The 2022 Pi Pico W is developed with onboard BLE capabilities. Software access to these features was unveiled in February of 2023 but has not yet been explored to its fullest capabilities. The final electrical system architecture is shown in Figure 4

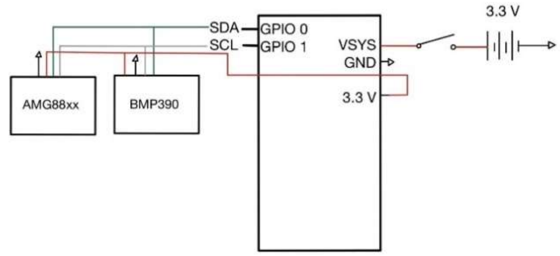


Figure 4 Sensor package circuit diagram.

2. Communications

Raspberry Pi's reputation as an accessible hobbyist-friendly microcontroller motivates electrical engineering efforts to explore communications. There are three options of wireless communication when working with the Raspberry Pi Pico W: Wi-Fi, Bluetooth, or Bluetooth Low Energy. Wi-Fi is difficult to implement due to restrictions at the center, as well as the complexity of setting up the connection. For Bluetooth, the Pi Pico W has both Bluetooth Classic and Bluetooth Low Energy (BLE) capabilities. However, the official support for Bluetooth on the Pi Pico W is fresh, and development documentation has not yet been released. BLE was used in the final iteration and can maintain connectivity up to 300 feet.

Serial Bluetooth Terminal is used to connect the Raspberry Pi Pico W and receive data transmissions. The application enables data transmission from the Raspberry Pi Pico to a user device. This is performed through a communication portal. Communication is done with line-by-line messages. The device verifies commands sent through the communication portal according to the program flow in Figure 5. Once commands are confirmed, a response relevant to the command is transferred to a user's device. This structure lets the user define multiple commands, allowing for different sets of data to be transferred without being strung together.

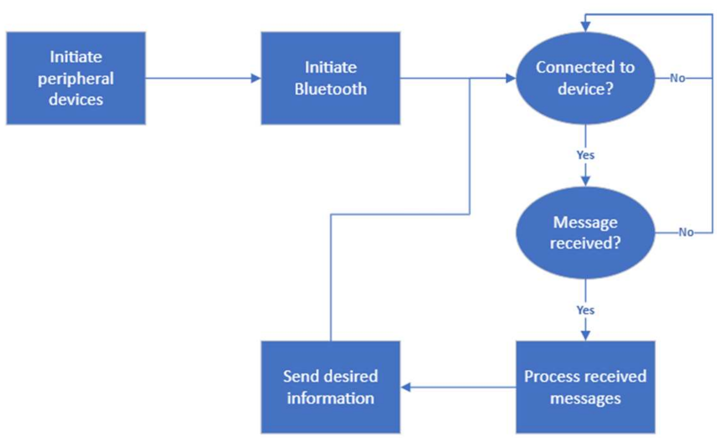


Figure 5 Flow diagram of Pi Pico W communications.

To develop an easily digestible user interface, LCD screens are considered. However, while prototyping, integration proved difficult. Between problems with ensuring a common language and accessing compatible files, LCD screens are confirmed to be more problematic than advantageous. In working with the LCD screen, it is hypothesized that the system would use 2 Pi Pico Ws to communicate with each other through Bluetooth. As the Bluetooth capabilities of the Pi Pico Ws are developed, the seamless interaction the system has with the Bluetooth Serial Terminal Application is discovered. It is concluded that a mobile device is more advantageous than developing a system-specific user device. The visualization of this Bluetooth architecture is shown in Figure 6.

Figure 6 Visualization of Bluetooth system architecture.

3. Reading Interpolation

On board the flight body, AMBG8833 matrices are married to altitudes found through the BMP390 Altimeter. These values are saved and iterated over such that matrices with similar altitudes are compared. This ensures an equal imaging resolution across flight. A third matrix is generated from comparing values in each matrix. This matrix distinguishes heat flow through comparison of perceived temperature values. By subtracting the two comparable matrices, the developed fire perimeter is detected. If imaging resolution is standardized, each position in the matrix can be generalized to a certain resolution. As a result, any change in each step of the matrix can be perceived as a velocity. The matrix of heat transfer direction is then transmitted to the ground operator. An example of this resultant direction matrix and its corresponding temperature matrices is shown in Figure 7.

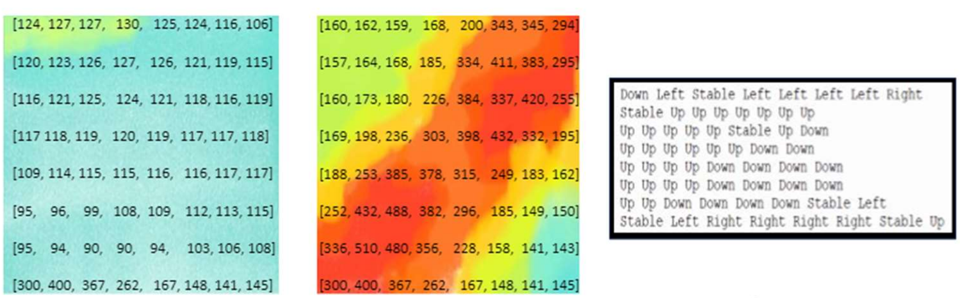


Image 1
@12:00:00 am

Image 2
@12:00:07 am

Resultant Matrix
@12:00:07 am

Figure 7 Example resultant and its corresponding temperature matrices.

In application to wildfires, a representation of this data is visualized in Figure 8. This demonstration shows the change in fire perimeter position as well as the direction it takes.

Image 1: 12:00:00 PM Image 2: 12:07:00 PM Difference

Figure 8 Application of generated matrices to wildfires.

B. Descent System

1. Parachute Canopy

The descent rate of a parachute is highly contingent on the sizing of the canopy diameter. With the relation between the descent speed and the canopy diameter, an equation calculating the terminal descent speed is used. With inputs of the system weight, W, based on the body and sensor payload taken to be 0.5 lbs, coefficient of drag, C_d, based on the parachute design configuration taken to be 1.5, and air density, ρ, taken to be 0.0767 lbs/ft^3, equation (1) shows parachute diameter as a function of desired descent speed [3].

$$d(V) = 76.95 * \sqrt{\frac{W}{V^2 C_d \rho \pi}} \tag{1}$$

For CLASSy, a 16-inch diameter parachute results in a theoretical descent speed of just under 15 feet per second, allowing ample time to both sense and transmit data to the user. With the use of model rocket parachute folding and packing techniques, a 16-inch parachute was also able to compactly fit within a 2-inch diameter body tube.

To aid in construction, the parachute canopy was split into sections, called gores. The number of gores in the parachute canopy represent the individual sections of a parachute. It is also found that stress in the parachute canopy is relieved by making the gore width at the vent 10% wider than the calculated gore dimension [4]. The number of gores is significant in maintaining the parachute shape during descent. For a hemispherical parachute canopy, a larger number of gores provides a more uniform hemispherical shape. However, this adds complexity in manufacturing and needs for additional material. For CLASSy, a gore number of 6 was chosen.

To ensure adequate data is acquired and transmitted, appropriate descent speeds and oscillation angle ranges are desired. In designing the parachute, a trade-off between this desired sink rate and stability exists. To combat this, additional parachute design alterations are considered. The use of apex venting, cutting a hole at the top of the parachute canopy allowing for turbulent air to escape, slightly increases the descent speed as well as increases stability. The apex vent for a non-reefed parachute should remain between 0.25% and 1% of the total canopy surface area [4, Ch. 7.3.9.1, pp. 7-30]. Additionally, a 1-inch free space should remain between the radials, the seams of each gore section of a parachute. In the case of a 16-inch parachute chosen for CLASSy, an apex vent area equal to 0.686% of the total canopy surface area, or a 2.76-inch diameter apex vent hole, fits within the constraints in maximizing parachute descent stability without drastically altering the descent rate. It is also found that the vent hole diameters should remain under 32.5% of the diameter of the parachute canopy to avoid failures in the inflation process of parachute deployment [5].

To counter the increased descent speed of the parachute from the use of the apex vent hole, vent reefing, a technique to alter the way a parachute opens, is explored. A vent-reefed parachute features a line pulling the apex towards the point of confluence, effectively vertically compressing the parachute's canopy. Vent reefing is commonly used in the design of annular parachutes to significantly increase the drag coefficient, ultimately decreasing the descent speed. The implementation of a drogue parachute to enhance descent stability is also in heavy consideration as it provides a descent speed high enough to prevent disruptive oscillatory behaviors and significant drift while still allowing for a main parachute to deploy. As implementing a drogue parachute in the sequence adds complexity in multi-stage deployment, for the case of CLASSy, the use of a drogue parachute is discouraged.

Another significant factor in the parachute construction that affects stability and descent speed is the parachute canopy shape. Low complexity in the construction of the parachute canopy is heavily emphasized as CLASSy is meant to serve as a low-cost expendable system. Simplicity in construction is also beneficial to both the manufacturer and user to ensure sufficient parachute packing into the body tube, ensuring successful parachute inflation when deployed. The first consideration in the parachute canopy shape is porosity. Parachute porosity is chosen based on the descent speeds that are experienced. A low-porosity solid parachute is recommended as CLASSy is designed to only experience subsonic descent speeds. Additionally, a solid parachute consists of multiple solid gores, simplifying the construction and packability for manufacturers and users.

The initial design of the parachute canopy is chosen to be a hemispherical parachute with a hemispherical gore shape. When unvented, this parachute shape has a coefficient of drag ranging from 0.7-0.9. Though a hemispherical parachute is known to suffer from oscillatory instabilities during descent, this is combatted through the use of apex venting. Due to the subsonic nature of CLASSy, this parachute canopy design is sufficient as the suspension elements would not experience heavy opening loads. This canopy design is also chosen as it is relatively low-cost to manufacture compared to other more complex parachute canopy geometries.

Additionally, annularity in CLASSy's parachute design is considered. Annular parachutes are traditionally used when a slow, steady descent speed is desired. Annular parachutes utilize reefing techniques coupled with comparatively large vent holes to perform their function. This parachute canopy shape has an increased coefficient of drag by increasing the nominal diameter to create a larger surface area [6].

For testing purposes, the parachute canopy is made of a ripstop nylon material. Ripstop nylon is conventionally used for modern parachutes as it has a favorable strength-to-weight ratio, is water and ignition resistant, and has minimal porosity. Additionally, ripstop nylon is relatively easy to access, which is advantageous for prototyping purposes. As CLASSy is meant to serve as an expendable, single use device, an environmentally safe material for the parachute canopy construction is investigated. Silk is recommended as the final design material as it is the strongest of all natural fibers, fully degradable, resists heat, and only burns when in direct contact with a flame [4, pp. 6-75].

2. Suspension Method

In lieu of conventional suspension lines, a mesh or netting material supports the payload for CLASSy's missions. Utilizing mesh suspension methods affords the design two important advantages. Seeing that this device is to be deployed in natural disasters in which the environment is volatile and unforgiving, the mesh design decreases the overall risk of entanglement, whether that be during packing, upon launch, or in flight. Additionally, the mesh suspension system continuously and evenly loads the skirt of the parachute increasing the general stability of the vehicle for the descent during which the appropriate data can be collected. The length of the suspension system is equal to the nominal diameter of the parachute, as recommended by the 1978 issue of the *Recovery System Design Guide* [7]. This places the confluence point of the suspension mesh in a desirable range and therefore minimizes unwanted oscillatory patterns. Due to the apex vent hole in the canopy, mesh material covers the opening, allowing the reefing line to attach to the true apex of the parachute while still allowing airflow through it. The elimination of easily tangled suspension lines obviates concern of tangling and ensures a smooth descent.

3. Parachute Testing

To determine and validate the final design configuration, drop tests are performed. In prototyping different design configurations, an initial test article is fabricated, using ripstop nylon for the canopy and meshing material for the suspension system. The procedures followed to accomplish prototyping can be found in Appendix D. Additionally, preliminary testing procedure validation trials are performed at a 25-foot drop wall to define the design parameters to manipulate, as well as the desired data to be collected from the final parachute drop tests.

As found in the final drop testing procedures in Appendix E, the two design parameters are the following: apex vent hole sizing and reefing line length. The larger vent holes allow more turbulent flow fields to escape the underside of the canopy, increasing stability. However, enlarging the vent hole compromises the projected canopy area, effectively decreasing the drag coefficient and increasing the sink rate. A shorter reefing line compresses the canopy, as it pulls the apex of the parachute towards the point of confluence, maximizing the projected canopy area. Theoretically, this increases the drag of the system. The three types of data to be recorded during the testing are the following: descent rate, general drift, and video analysis for the oscillation of the parachute. The descent rate provides how much hangtime the design allots for the sensor system to gather the required data. The general drift assesses the sensitivity to the wind environment. The video analysis characterizes the oscillatory behavior of the parachute-sensor system.

As discussed in the results of these drop tests in Appendix F, the design featuring an apex vent hole whose area is 0.686% of the total surface area of the parachute canopy with no reefing line is the optimal parachute configuration. Based on data from Table F-1 and Equation 1, this design results in an average descent time of 10.72 ft/s and a coefficient of drag of 1.831.

4. Parachute Deployment

For the final design, the parachute's deployment is a completely passive process. Upon reaching apogee, a CLASSy sensor system inverts in the air, allowing the nose cone to detach from the rest of the body. The ventilation of air through the body as it begins to fall forces the parachute to eject and inflate. The parachute is connected to a shock cord which allows the parachute and body to move separately during the descent, allowing stability for on-board data-collecting devices.

C. Body Design

1. General Geometry

A model rocket best satisfies CLASSy's design constraints. In researching, it is found that there already exist model rockets with similar use cases to that of CLASSy's. Therefore, it is advantageous for prototyping purposes to modify an existing payload-carrying rocket. This ensures the rocket is stable and aerodynamic prior to any design modifications. The existing model chosen is the Majestic Model Rocket Kit, Pro Series II E2X [8]. This original design is around 35 inches tall, has a diameter of two inches, and has three trapezoidal fins. Modifications are made to change the height to 19 inches and an additional fin was added. The body is made from 0.2-inch-thick cardboard. The total weight of the rocket is approximately 1 pound when ready for launch. Although a larger rocket may be able to reach increased heights, it would be far less mobile for ground personnel. Consequently, the body must be as small and as lightweight as possible, while still being able to reach the desired launch height. The necessary diameter of the rocket is determined by the size of the folded parachute configuration. If the diameter of the rocket is smaller, there will be too much friction between the parachute and body tube, causing a failure when deploying the parachute. A shock chord is attached to the nose cone and the body to keep the two components together after the parachute has deployed. The shock cord is made of elastic which helps absorb the shock upon separation. The overall stability of the rocket after modifications is confirmed through OpenRocket [9], and these calculations can be found in the Appendix G. This model shows the parachute being 1.5 inches inside the nose cone and the payload as 6.75 inches away from the base of nose cone. The fins are also 1.5 inches from the base of the rocket. This makes the center of gravity 12.318 inches and the center of pressure 14.021 inches from the tip of the rocket.

2. Aerodynamics

The overall aerodynamics of the rocket is a critical aspect of design. The body needs to be able to "cut" through the air using the nose cone, the trajectory needs to be straight and stable through the fins, and it needs to weigh just enough to not be consumed by air drag before reaching apogee. Additionally, the center of gravity (CG) needs to be in front of the center of pressure (CP) to ensure the body does not flip around in flight. The center of gravity is the point where the weight is balanced, or where there is an equal amount of weight fore and aft of the point [10]. The rocket will always spin around the CG point. The center of pressure is the point where the aerodynamic forces are balanced, or where there is an equal amount of lift and drag fore and aft of the point [11]. The relationship between the CG and the CP defines the stability of the rocket. The further forward the CG is from the CP point, the more stable the rocket becomes [12]. To help stability, modifications can be made to the nose cones, the type and number of fins, and variations in body diameter. For CLASSy, the nose cone and fins are modified.

3. Nose Cone

The shape and material of the nose cone is determined through research on aerodynamic performance, manufacturability, and the contribution to weight. To optimize aerodynamics and ensure the air flows smoothly over the surface, the shape needs to have minimal sharp edges or abrupt changes in geometry [13]. This is important as it prevents flow separation, which causes drag. As a result, a steep conical shape is chosen, where the circular base has a diameter of 2 inches. The total height is 5 inches, where 1 inch of that is a shoulder that allows the nose cone to sit inside of the body. The protruding height is then 4 inches, making the total height of the rocket 19 inches. The nose cone is made from a lightweight foam that is carved to the desired shape. As a result, the finish is rough and has many imperfections which greatly impacts the aerodynamics. However, this was made for prototyping purposes and is not intended for the final design. Another prototype is made from balsawood that is slightly heavier, but still fulfills the requirements set by the team. The material being balsawood also means it is environmentally friendly, cheap, and easy to acquire. The nose cone being slightly heavier is also not always detrimental. Adding weight to the nose cone helps move the CG forward and away from the CP. However, this is not always the most efficient way to separate the CG and CP as adding unnecessary weight is not ideal.

4. Fins

One of the biggest contributors to rocket stability is the number and placement of fins. Increasing the number of fins moves the CP back and further away from the CG [14]. As a result of this, four fins are chosen to be placed near the base. The type of fins is also heavily considered when making the final design. It is ideal to have fins with ample surface area to redirect the flow; therefore, multiple fin designs are researched. In addition to aerodynamic stability, manufacturability is another factor that impacts the decision. As a result, the design is narrowed down to clipped delta and trapezoidal. Both fin shapes have good aerodynamics, but trapezoidal fins are mainly used for payload-carrying rockets. However, they also move the CP forward, which is not beneficial to a shorter rocket such as the model chosen. The manufacturability of the clipped delta shape is better and kept the CP and CG further apart, so it is decided that this is the final design. The fins were made from basswood, which proves to be more durable than balsawood and provides greater flutter prevention. Finally, the edges of the fins are sanded along the leading edge to be shaped as airfoils.

D. Launch Mechanism

1. Launch Method

To begin development of the CLASSy launch mechanism, the launch team prioritizes requirements that are deemed most important to first responders. These requirements and their justification can be viewed in Table 5. A trade study, found in Appendix H, is performed in which a variety of launch methods are researched. The use of solid rocket motors (SRM) or other flammable/explosive propellants are dismissed due to the preestablished requirements. The results of the trade study identify elastics to be the most plausible solution out of the propulsion methods. Elastics provide the most flexibility in design since they can be changed based on size, strength, and durability.

Once elastics are selected, external factors that might influence the design constraints of the launch mechanism are focused on. Regarding launch, the NASA Langley Research Center has set height restrictions that limit the maximum allowable launch to 200 feet. For the purposes of on-site testing, 200 feet is selected as a design constraint. The prototype is designed and developed on-site, which also introduces fabrication limits. The first prototype is constructed primarily of 3D printed PLA filament. This material is selected due to the ease of manufacturability, accessibility, and the versatile nature of 3D printing. For details on the role of 3D printing in the development model, see Appendix K.5.

Conclusions from the design process determine that elastic potential energy is most suitable for this launch mechanism application. Simplified estimate calculations, found in Appendix I, are conducted on the elastic potential energy required, and compression springs initially investigated. The method of storing and releasing that potential energy was the next concern. Multiple architectures are considered for how to go about this, but ultimately a notched raceway and keyed slider are chosen for its mechanical simplicity.

While conducting rough calculations for the elastic spring constant, k, required to achieve our target altitude, it is found that no springs are readily available at that k value that are also light, compact, and inexpensive enough to match our other system requirements. After much research and estimate energy calculations, it is determined that elastic rubber material must replace springs as the potential energy source for this system [Appendix I]. Low-strength rubber bands are initially used to prove the functionality of the development model in a safe way before moving to high-strength elastic material. Using a load cell, several rubber bands of varying unstretched radii and thickness are tested to look at material fatigue and determine k constants. For details on experiment descriptions, testing procedures, and resultant data, see Appendix J.

Research is conducted on what high-strength elastic material is required for an engineering model of the launch mechanism that would store the potential energy needed to reach the target launch height. It is discovered that high resistance tubing is ideal for this application in terms of strength and elasticity. Once the resistance tubing is acquired, additional experimentation and testing is completed to determine the k value of the new material. The results, which can be found in Appendix J, yielded a favorable outcome in the selection of the material to be tested in the final development model and ultimately used in the high-fidelity engineering model.

2. Development Model Design

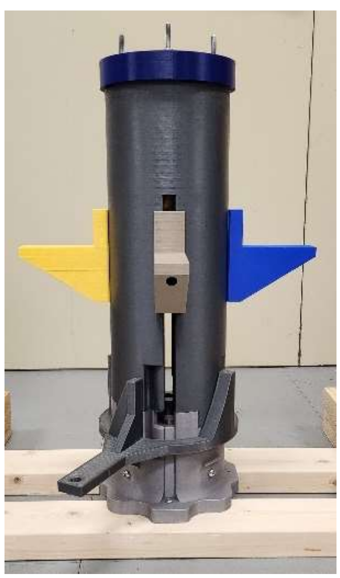
Based on initial energy and force calculations for launching a payload of approximately 0.5 pounds to 200 feet, it is determined that the cocking and locking of the potential energy needs to be distributed across four independently cocked and locked raceways. This decision is made to meet the driving requirement of a single adult operating the mechanism. The shape of the full prototype architecture resembles that of a mortar tube with dimensioning matched to size constraints. The mechanism is attached to the ground using simple stakes. Foot pedals are directly attached to the sliders of the raceways and are used to ergonomically cock the elastic potential energy. For a final high-fidelity prototype, it is estimated that the load to cock an individual keyed foot-pedal is no more than 100 pounds of force. A cam-collar with an attached lever is used to synchronize the release of the keyed foot-pedals in their notches, allowing them to freely ride up their respective raceways. For a detailed diagram and further description of this system architecture, see Appendix K.1. For details on the CAD methods used to design the parts for the development model, see Appendix K.3.

A proof-of-concept prototype is designed and assembled to demonstrate the notched raceway and keyed slider in a 3-stage cock, lock, and release mechanism for a single raceway using low-strength compression springs as the potential energy source. This mechanism is developed in tandem with the notched raceway concept. For details on the development of the notched slider and keyed raceway architecture for this cock, lock, and release mechanism, see Appendix K.2. Once it is proven that the mechanism is viable, the team moves forward with prototype development.

The foot pedal sliders, being 3D printed out of PLA, are susceptible to failure at the joint connecting the pedal with the raceway slider. Hardware and new printing methods were introduced to strengthen this part, and impact testing was conducted to observe the viability of these changes; see Appendix K.4 for details.

3. Prototype Manufacturing

The fully assembled model, with low-strength rubber band elastic material and 2x4 wood frame connected to base for testing stability, see Figures 9 and 10, has a height of 23.25 inches with a 6.5-inch outer diameter. Basic hardware in the system includes washers, eyebolts, carabiners, and simple 1/4 - 20 screws and bolts. It weighs approximately 9 pounds. A base frame is constructed out of 2x4 wood studs to screw the base of the 3D-printed assembly onto, providing the model with stability when testing. Hinges and ground stakes are recommended to stabilize the model for launch. See Appendix K.6 for detailed screenshots of the development model CAD assembly.



Figures 9 & 10 Full-scale development model assembly, low-strength rubber band elastic material, 2x4 wood frame connected to base for testing stability & Exploded view of development model.

The prototype is built to demonstrate the individual cocking of each of the four-foot pedals and the simultaneous release mechanism. It is initially tested using the low-strength rubber bands to ensure the mechanism works well. To ensure a safe and proper launch, a Pre-Launch Checklist and Launch Procedure was created. This checklist can be viewed in Appendix L. With rubber bands the maximum launch altitude observed in testing was 15 feet with a theoretical calculation of 20 feet. The 25 percent discrepancy can be attributed to frictional energy losses between the PLA-on-PLA contact of the foot pedal sliders with their respective raceways as well as destabilization in launch.

Once the mechanism was deemed functional, high-strength elastic resistance tubing replaced the rubber bands as the potential energy source for further testing. The highest launch of the payload body with the elastic resistance tubing is 30 feet, which was close to the calculated theoretical height, assuming ideal energy transfer conditions, of 40 feet [Appendix I].

IV. Future

As an abbreviated engineering solution to the problem of last-mile connectivity, CLASSy's design features many areas of improvement due to the condensed fabrication timeline of only 10 weeks. From ideation to prototyping to testing and verification, CLASSy has forgone several fundamental milestones that would allow for more robust and precise design decisions. Consequently, towards the end of this project, there have been many flaws that would have been accounted for in the initial designs and prototypes, had adequate time been allotted for research on the subject-matter and for consultation with experts.

In reference to the electrical system, development is to be centered around improving the user interface as well as imaging techniques. Next steps for the sensor unit entail the development of vital landmark recognition and implementation of machine learning, more specifically higher-level intelligent systems. As predictive computations gain more accuracy and precision, AI will assist in adding heighted levels of confidence to frontline decision support. However, a tradeoff exists between the integration of AI along with the cost and weight of CLASSy. The current state of the art technology, when it comes to advanced intelligent technologies, feature heavier and more costly equipment. Additionally, the design of an independent user interface separate from the Bluetooth Serial Terminal application would be advantageous in providing a more seamless user experience.

In refining the parachute for CLASSy's descent, alternative materials would be explored with emphasis on environmentally friendly design. As another consideration for material selection, integration with the body would be important, ensuring that the parachute would be able to fold and fit within the fuselage while still being able to slip upon parachute deployment. A major problem with the current prototypes is that the skirts of the canopies are rigid, making any reefing efforts useless, as the projected area of the canopy remains unchanged. An ideal material would be slightly elastic, enabling vent reefing techniques as a means of decreasing the average sink rate of the parachute.

Going forward with body frame development, the use of accurate machining techniques to manufacture parts such as the nose cone or fins is desired. This would prevent any possible human error in the machining process and would ensure each part is as symmetrical, and therefore as aerodynamic, as possible. The use of a hollow nose cone is also of benefit to conduct further research on. A hollow nose cone would lower the total weight of the body frame and create more space for parachute and payload housing within the main tube. Additionally, different methods of parachute deployment should be explored. Although passive deployment keeps CLASSy low-tech and low-cost, there is the possibility that the nose cone may not always fall out at apogee which would be catastrophic for CLASSy's mission. Therefore, other feasible methods that do not interfere with the harsh environment of wildfires (i.e. using an explosive charge) are valuable to consider.

The final launch mechanism engineering model design better fulfills the system requirements and design constraints, specifically the weight and launch height. The architecture an outer shell, four foot pedal sliders contained in raceways attached to high-strength elastic tubing, and removable top plate and housing base for servicing designed in the development model is maintained in recommendation for the engineering model. New implementations to reduce energy loss through friction between parts should provide the kinetic energy required for desired apogee. The locking mechanism is also maintained, using the notched raceway and keyed slider architecture system that was developed and previously described.

The team recommends that the main tube of the final mechanism consist of a graphite-honeycomb composite. This composite is three times lighter than PLA, reducing mechanism weight beneath the 5-pound design constraint. The composite has stronger shear and impact thresholds, theoretically allowing it to withstand the loading and high-impact forces within the structure.

Aluminum hardware is integrated to reduce the friction of the sliders while allowing the weight budget to be maintained. 80-20 aluminum framing extrusions are used as raceways for specified slider hardware to interface with a cast or 3D printed foot pedal and elastic bands. These extrusions are inserted into a custom modeled baseplate used to both attach the outer tube and keep the extrusions in place; this part can also be 3D printed as it will be non-load bearing. Since these proposed raceways are square, a new slider architecture is recommended to maintain the notched raceway and keyed slider lock and release architecture. This assembly consists of Teflon washers, tapered thrust roller ball bearings, and aluminum ring clamps. For more details and images of each recommended part for the engineering model, view Appendix M.

Due to the viability of resistance tubing over rubber bands, they are recommended for use in the engineering model. To decrease the torque moment between the foot pedal and aluminum extrusion so the sliders do not bind in their respective raceways, each slider has two bands attached to it, each on opposite sides of the framing extrusion. The resistance tubing and the graphite honeycomb composite collaboration fulfills the crucial design requirements. Given that the final assembly was not tested, this section is purely recommendation for future development.

V. Conclusions

CLASSy is a low-tech, cost-effective disaster mitigation tool that provides decision support for firefighters based on fire perimeter detection from an aerial point of view. It combines an all mechanical, elastic based launch system with an IR camera sensor, descent system, and model rocket body. Through this collaboration, CLASSy revolutionizes active disaster mitigation through technological addition. CLASSy balances the advantages of modern technology and the utility of disposable, mechanical systems. With the developments taken, wildfire relief operation has been streamlined to efficiency.

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Appendix A: Alternative Mission ConOps

To better visualize the plethora of use cases for CLASSy, a series of alternative mission ConOps was created. The alternative use cases are shown in Figure A-1 and highlight post flood, hurricane, tsunami, avalanche, and train derailment relief. These ConOps show how CLASSy can be used for a variety of search and rescue missions as well as environmental data collection during times of poor air quality.

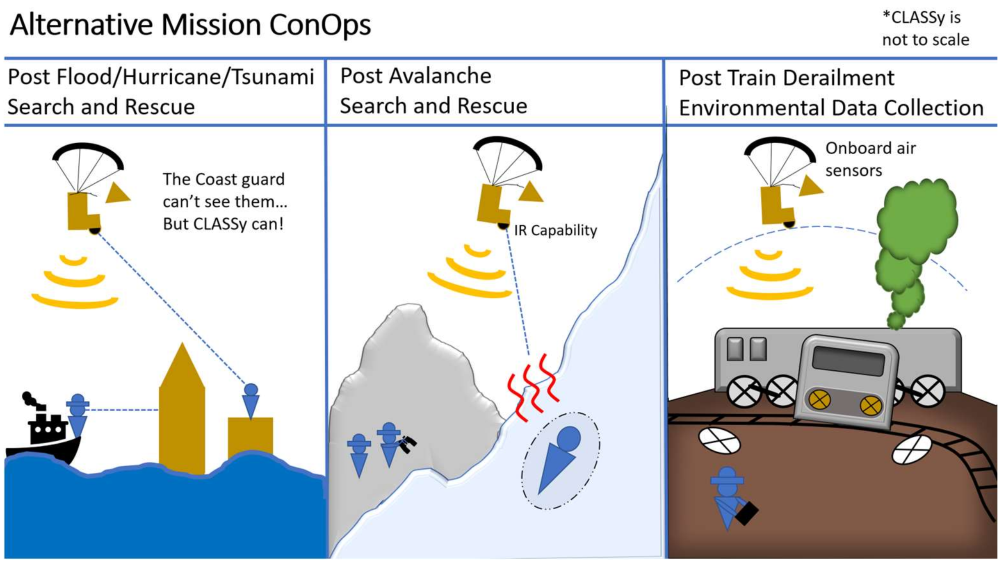

Figure A-1 Alternative mission concept of operations.

Appendix B: Lifting Mechanism Trade Study

A more refined trade study explored the use of either an auto-gyro, kite, balloon, or parachute to support CLASSy's mission. This trade study separates system requirements by wants versus needs. Each alternative is ranked on their satisfaction of each requirement on a "stop-light" scale where green means fully meets, yellow partially meets, and red does not meet the requirement. Figure B-1 shows the alternative with their associated rankings based on requirement.

			Alternatives			
		Requirements	Auto-Gyro	Kite	Balloon	Parachute
NEEDS	R1	Firefighters can operate without removing personal safety equipment	2	2	2	2
	R2	Total aeriel mechanism weight is <4.4 lbs	2	1	3	2
	R3	Mechanism is reusable for multiple missions (minimum 10 deployment lifetime)	1	2	2	2
	R3	Mechanism can be launched (deployed) by ground crew in <3 minutes	1	1	1	1
	R5	Mechanism has module sensor package housing	1	1	1	1
	R6	Prototype can be manufactured using Langley resources	1	1	2	2
	R7	Prototype materials to make mechanism are readily available at Langely	1	2	2	2
	R8	Academy can Design/Prototype/Proof of concept test in ~7 weeks	2	1	2	1
	R9	Mechanism maximum length is 1 ft when compacted	1	1	2	2
	R10	Maximum housing diameter is 3 in when compacted	1	1	2	1
	R11	Minimum housing unit for the sensors is 2 in^3	1	1	1	1
	R12	Mechanism must not exceed 400 ft AGL with tether	1	1	1	1
	R13	Must be able to exit air space quickly (< 2 minutes to treetop canopy)	1	1	2	1
	R14	Sensor package data accessible to ground crew (<2 minutes to access data files once grounded)	1	1	1	1
	R15	sensor package orientation stability in high-wind conditions (40 mph winds)	1	2	3	3
	R16	High heat transfer resilience (safe operation up to 500 degF)	1	1	2	2
	R17	Impact damage resilience (mechanism maintains deployablility after 30 ft freefall impact)	2	2	3	2
	R18	Final product lifetime cost (< $1,000)	1	1	2	2
	R19	Low cost to prototype (< $500)	2	1	2	2
WANTS	R20	Mechanism capable of self-propelled ground launch	1	1	1	2
	R21	Mechanism is passively powered for flight/hover operation	1	1	1	1
	R22	Mechanism is module for various disaster relief scenarios	1	1	1	1
	R23	Low power requirements for launching	2	1	2	2
	R24	Low power requirements for sensor package	1	1	1	1
	R25	Ground operator has manual steering control capabilities via tether	2	2	2	2
	R26	Ground launch requires small (< 10 ft^2) treetop canopy area clearance	1	3	2	1
	R27	Automated tether reeling to reduce ground crew involvement	2	2	2	2

Figure B-1 Trade study with refined options.

The results of the rankings were separated into pie charts to demonstrate the amount that each alternative fully satisfies the requirements. One set of pie charts show the rankings for both system wants and needs while another set of pie charts show rankings based only on system needs. These pie charts are shown in Figures B-2 and B-3 respectively.

Figure B-2 Results summarized for wanted and needed requirements.

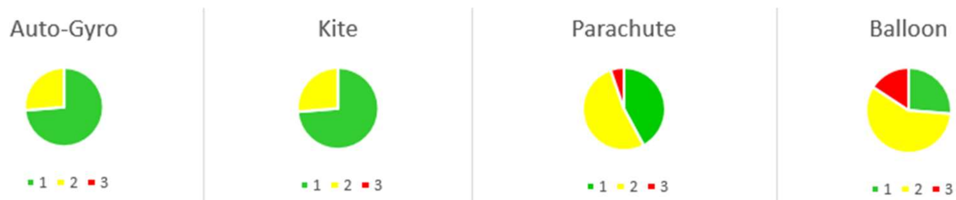

Figure B-3 Results summarized for only needed requirements.

Appendix C: Electrical Component Trade Study

Table C-1 summarizes a trade study performed on the different components of the sensor unit based on cost, data resolution, durability, and adaptability requirements. Rankings of each component for each feature are represented as either green, yellow, or red. Where green is advantageous, yellow raises caution, and red is non-desirable for the system.

Table C-1 Summary of potential electrical system components.

	Component				
Feature	AMG8833 IR Camera Breakout	MLX90640 IR Array Breakout	BMP390 Pressure and Temperature Sensor	MCP9808 High-Accuracy Temperature Sensor	Raspberry Pi Pico W
Total Cost <$100	$44.95	$74.95	$10.95	$4.95	$7.00
Data Resolution	8’x8’x20’ Matrix of Celsius Values	55 by 35 degrees of view Matrix of Celsius Values	Float Values	Float Values	Bluetooth LE connection up to 300 ft
Durability	Withstood Impact test	Failed Impact test	Withstood Impact test	Withstood Impact test	Withstood Impact test
Adaptability	I2C Communications Versatile Formatting and Integration	I2C Communications Versatile Formatting and Integration	I2C Communications Versatile Formatting and Integration	I2C Communications Single Integrative Use	I2C Communications Adaptable Programming Language

Appendix D: Parachute Fabrication Procedures

1. Parachute Canopy
   a. Calculate required canopy diameter, gore number, and choose canopy design
   b. Draw and cut out flat gore template out of foamboard
   c. Trace and cut out flat gores out of ripstop nylon
   d. Connect the gores using ripstop nylon tape
   e. Trace and cut out parachute vent
   f. Sew mesh lining on vent hole

Construction of the parachute started with calculating the required canopy diameter, gore number, and canopy design to match the desired descent velocity. To aid in cutting out the parachute canopy out of ripstop nylon, a stencil was cut out of foam board after an outline of the parachute gores were calculated and drawn. With the stencil cut out of the foamboard, the gore shape was then traced onto ripstop nylon and cut out. With the gore sections cut out, each were connected using an adhesive ripstop nylon tape. Once the canopy was formed, an apex vent was calculated, traced, and cut out of the top of the canopy. To cover the apex vent, a mesh lining was sewed to allow for airflow while covering the hole to prevent any tangling.

2. Parachute Suspension
   a. Calculate the mesh suspension sizing
   b. Draw and cut out mesh suspension line shape out of foamboard
   c. Trace and cut out suspension line shape in mesh
   d. Sew mesh suspension together

Construction of the mesh suspension started with sizing to have a length equal to the parachute canopy diameter. Again, to aid in cutting out the parachute suspension, a stencil was cut out of foam board after an outline was drawn. Using the stencil as a guide for a knife, the mesh suspension was cut out. Each of the sections were then sewn together using a thread and needle.

3. Parachute Construction
   a. Line bottom of canopy with adhesive tape
   b. Sew mesh suspension to canopy

To combine the two sections of the parachute, the bottom edge of the canopy was first lined with ripstop nylon tape to prevent fraying of the ripstop nylon. The mesh suspension was then sewed to the parachute canopy. To test reefing in the parachute design, a single thread was tied onto the apex vent mesh and pulled down to the bottom of the mesh suspension.

Appendix E: Drop Testing Procedures

Design Parameters

The two primary design parameters to be manipulated during parachute development will be the size of the apex vent and the reefing line length. Different sizing-length combination performances will be observed and quantified through a series of repeated drop tests.

Design-Dependent Variables

The three qualities of the drop test that will be measured and discussed in this testing will be the following: general drift, terminal sink rate, and oscillatory behavior of the payload. The general drift and oscillatory behavior will allow for commentary on the stability of the parachute configuration and design. The sink rate will demonstrate hangtime, which corresponds to the window the device must collect data.

Required Environmental Measurements

The environment can dramatically affect the trajectory of parachute systems. Measurements required to appropriately assess data collected during these drop tests will be the wind direction and speed at or near the drop height and the general wind direction (and speed if significant) at ground level. These values will inform discussion related to the drift patterns.

Landing and Impact Research Facility (LandIR) Testing Site

This test will be performed at the LandIR on center at Langley Research Center. Drop tests were performed from the 150- and 200-foot platforms.

Safety Protocols

To avoid unpredictable descent trajectories, calm weather (<10 mph winds) was required. Hard hats were always worn while testing was underway under the gantry. Additionally, for those who climbed the structure, they avoided leaning on railings. The landing site was cleared before each drop test was performed.

Test Procedures

1. Favorable testing conditions were acquired.
2. A camera was positioned in such a way as to capture a significant portion of the parachute predicted flight path in frame for later stability image analysis.
3. After donning the appropriate PPE, a parachute equipped with a sensor package capable of recording altitude along with time was brought up to drop height.
4. A last-minute check of the parachute-sensor assembly was performed to deter the creation of test article debris on the gantry floor. All joints and materials were visually inspected for obvious failures.
5. Coordinating with a ground crew, the test article was positioned for the drop, manually suspended only by the apex of the parachute.
6. The operator activated the sensor package, initializing the data collection of the article’s altimeter. All other recording devices were activated at this time.
7. The time and wind velocity atop the gantry structure were recorded as the required environmental measurements.
8. Simultaneously, the test article was released.

9. Once the test article lands within a target area under the Gantry structure, all recording devices were paused, and data was saved accordingly. Additionally, the sensor was connected to a computer, and the altitude and elapsed hangtime data for were stored.
10. This process was repeated three times for each decided design configuration.

Appendix F: Parachute Drop Test Results

This appendix officially documents the results from the testing performed at the Landing and Impact Research facility at Langley Research Center in Hampton, Virginia. Four parachute design configurations were tested. Configuration 1 had two test trials, while the rest were tested with one trial. Table F-1 describes the test configurations and displays the results and additional information concerning the drop test. Due to wind patterns at the time of testing, all drops were performed at the northeast corner of the gantry.

Table F-1 Design configuration descriptions and drop test results.

Drop Test Name	Drop Height, [ft]	Parachute Canopy Size, [inch]	Vent Hole Diameter, [inch]	Reefing Line Length, [inch]	General Drift, [ft]	Descent Duration, [sec]	Average Descent Rate, [ft/s]
Config. 1, Trial 1	200	16	3 (18.75%)	17.5	220	18.05	11.08
Config. 1, Trial 2	150	16	3 (18.75%)	17.5	115	13.11	11.44
Config. 2	150	16	1.88 (11.7%)	No Reefing	130	13.99	10.72
Config. 3	150	16	5 (31.25%)	No Reefing	157	11	13.64
Config. 4	150	16	3 (18.75%)	6	- -	7.1	21.13

Drop Test Observations and Comments

For each drop test, an observer recorded the parachute's descent trajectory as steadily as possible with a mobile camera. The following observations were gleaned from these video recordings of each drop test. These observations provide insight to characterize the behavior of each parachute as well as contextualize the data relative to wind data provided by an on-gantry weather station.

The first trial for design Configuration 1 was performed from a drop height of 200 feet atop the gantry. The average wind speed was 8 mph with a NE direction, tending north. Upon dropping, the parachute-sensor system drifted faster than expected, hitting a longitudinal support beam on a gantry walkway bridge. Although this did not drastically alter its average sink rate, it was determined that a lower drop height of 150 feet would be best for the purposes of the remaining parachute drop tests.

The second trial for Configuration 1 was therefore performed at a height of 150 feet. The average wind speed was 8 mph with a primarily NE direction. It was noted that the wind was much more favorable for this drop test than the last. During the first half of the drop, the parachute-sensor system experienced significant oscillations. As the test article descended, it gyrated in a circular motion. Around 30 feet from landing, the parachute encountered a brief stall in its descent before landing on the sidewalk that runs under the gantry.

The remaining configurations were all dropped at a 150-foot drop height and only had one trial each. The average wind speed for Configuration 2's drop test was around 12 mph. The wind direction was a bit more volatile, starting with a NE direction and shifting to flow directly east. At the beginning of this configuration's drop test, it descended calmly with minimal oscillatory behavior. After a few seconds, it begun to sway, but overall, the parachute-sensor system stayed relatively vertical.

The wind speed during Configuration 3's drop test was around 14 mph, consistently pointing east-northeast. The test article experienced many sudden changes in horizontal position, evidencing a much more volatile wind environment than in previous drop tests. This configuration seemed to be much more sensitive to the wind; however, this could be due to the heightened wind speeds.

Configuration 4's drop test did not go as expected. The canopy failed to inflate, resulting in squidding. The drop data from this trial was therefore determined to be insufficient to analyze or ascertain valid conclusions.

Conclusions

Based on its controlled descent and relatively minimal drift distance despite the elevated wind conditions, Configuration 2 was chosen as the optimal final design.

This was somewhat unexpected. Initially, it was theorized that a shorter reefing line length and a vent hole with a diameter of less than 30% of the canopy's nominal diameter would result in the best parachute drop test results. However, it was found that Configuration 2, which does not feature a reefing line, performed the best. The design of the reefing line was most likely to blame for this discrepancy between expectation and result. The thought process of compressing the canopy was to increase the projected area of the parachute, therefore decreasing sink rate. This would only work if the skirt of the canopy would allow for expansion. The prototypes used in this testing did not have much elasticity to the material in addition to be sewed in place. That said, the warped canopy most likely made it difficult for turbulent air to escape through the vent hole, disrupting its ability to stay aloft stably. Further investigation into designing the reefing system of the parachute would be required for a more robust assessment of its necessity for this use case.

To enhance the testing procedures for future drop tests, additional test articles would have been prepared with a larger variety in vent hole sizing and reef line lengths; descent tendencies would become more apparent. To ensure comparable testing environments, multiple test articles would be released simultaneously. Finally, more trials for each design configuration would be executed to eliminate discrepancy in the collected data.

Appendix G: Body Model

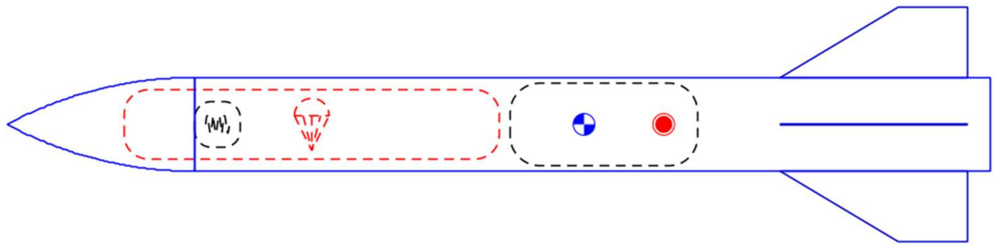

Figure G-1 OpenRocket Diagram of CLASSy Assembly

Above is the OpenRocket model of the CLASSy rocket. The model shows the location of the parachute in red, the shock cord next to the parachute, and the payload below the parachute. OpenRocket automatically calculates the center of gravity (blue circle) and the center of pressure (red circle). This model shows the center of pressure behind the center of gravity which is necessary for stability.

Appendix H: Propulsion Methods Trade Study

			Alternatives				
	Requirements		Elastic Cord	Elastic Spring	SRM	Compressed Air	Powered Ascent
NEEDS	RL1	Shall be operable by 1 operator	3	3	3	3	3
	RL2	Shall be deployable in under 5 minutes	3	3	3	2	3
	RL3	Shall energize sensor package to target altitude 200 feet AGL	2	2	3	2	3
	RL4	Shall be capable of being transported by 1 person	3	3	3	2	2
	RL5	Shall consist of non-hazardous material to humans	3	3	1	2	2
	RL6	Shall weigh less than 5 lbs.	3	3	3	1	2
WANTS	RL7	Cost less than $150	3	2	3	3	2
		TOTAL	20	19	19	15	17

Table H-1 Trade Study on Propulsion Method

Above is the trade study that was conducted in the early weeks of the project by the Launch Mechanism sub-team. It consisted of the 5 ideas that the team deemed most plausible to energize the sensor package to the desired height. The study was conducted between the determined requirements and the launch ideas. The boxes in red do not meet the requirement, those in yellow partially meet the requirement, and those in green completely meet the requirement. Numbers 1 to 3 were added to better visualize which launch method best fulfilled the requirements. Methods that had red boxes in the column were discarded from the discussion, as the team would rather have multiple requirements partially met than one requirement not fulfilled at all.

Appendix I: Elastic Potential Energy Theoretical Calculations

The math conducted to determine what spring constant, K, was required for the elastic that would store and release the potential energy needed to achieve the target launch height is as follows (first considered springs but then moved to elastic rubber material).

First, an initial velocity was calculated using a target launch height, denoted by h, with the basic kinematic equation: $V_f^2 = V_i^2 + 2gh$, where V_f was set equal to 0 for the apogee of the payload projectile's trajectory, and g set as the acceleration due to gravity. The initial velocity, V_i, could then be determined as a function of launch height. This initial velocity was used in conjunction with the mass of the payload, m, to determine an estimate for the kinetic energy required to launch to the target height using $KE = \frac{1}{2}mV_i^2$. The equation for elastic potential energy, $PE_e = \frac{1}{2}kx^2$ (where k is the elastic constant and x is the displacement length of stretching), was rearranged to set $PE_e = KE$ and solve for k. The resulting equation is $k = \frac{2KE}{x^2}$. Using an excel spreadsheet, we were able to create a vector of displacements, having the equation solved as a function of x, where the target launch height and payload mass were input as constants.

Once we determined from this spreadsheet and NASA vendors that there was no spring readily available that met our design criteria and had a k large enough, we applied the reverse of the above methodology to determine the available launch height from the tested elastic material. Appendix J provides how we were able to determine the k for a given elastic material.

Appendix J: Elastic Band and Resistance Tube Testing

Below is a series of graphs that display the results gathered from our elastic band and resistance tube testing. The initial loading tests were conducted using rubber bands of varying fatigue and age-induced brittleness that were given to the team. An experiment was conducted to compare the quality of bands to their respective strengths.

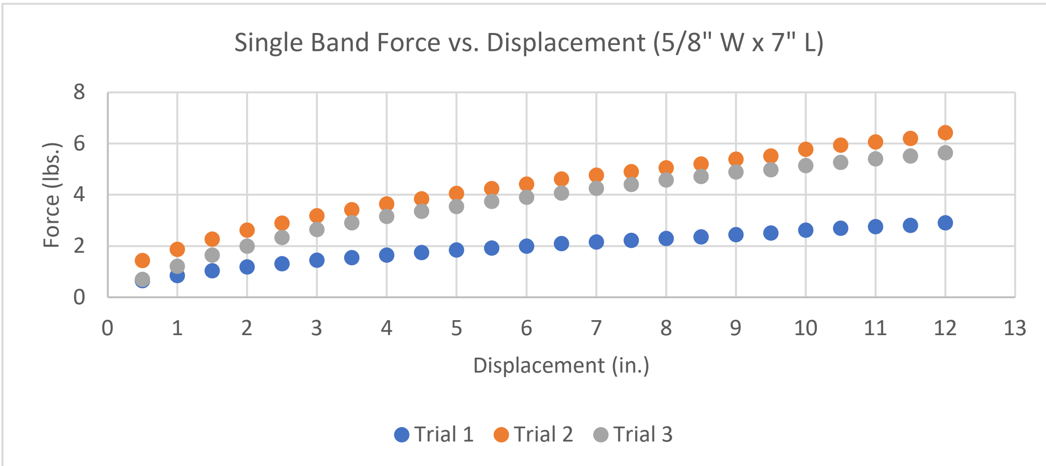

Figure J-1 The Force versus Displacement for a Natural Rubber band with an unknown age (7" L x 5/8" W)

Figure J-1 displays the force and displacement results from the first set of experimental trials. These natural rubber bands measured 7 inches in length with a thickness of 5/8 inch. The rubber bands used in this test were found around the office, therefore the age of the bands, along with their usage, was unknown. This provided a good basis to understand what to expect when testing proceeded with new rubber bands of the same dimensions. Based on the results above, these bands provided unreliable data and major discrepancies.

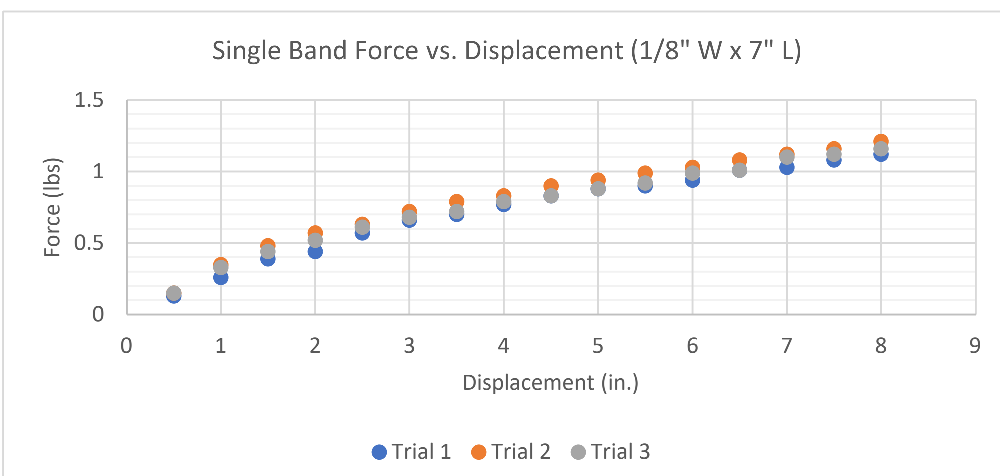

Figure J-22 The Force versus Displacement for a Natural Rubber band (7" L x 1/8" W)

Figure J-2 displays the force and displacement results from one of the experimental trials. These natural rubber bands measured 7 inches in length with a thickness of 1/8 inch. The bands used in this trial, and the remaining trials below, were brand new. To conduct the testing, the team looped one end of the rubber band to the leg of a stool and the other end was looped through a luggage scale. To provide additional support, an individual sat on the stool to prevent tipping. To test, the rubber band was stretched in increments of 0.5 inches and the luggage scale provided the force being produced. The displacement and the force produced at that given displacement was noted in an Excel spreadsheet. From this data, the graphs were derived. The new rubber bands provided reliable data and displayed continuity among multiple different bands.

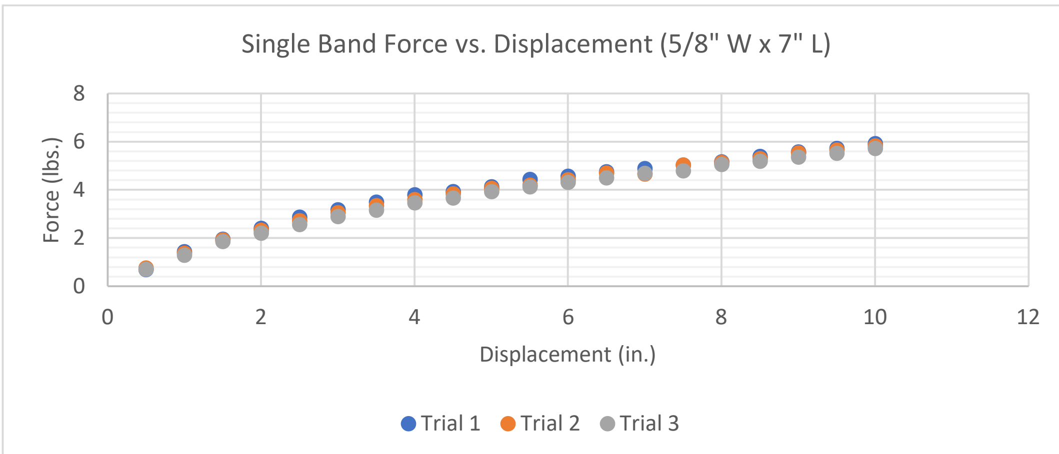

Figure J-33 The Force versus Displacement for a Natural Rubber band (7” L x 5/8” W)

Figure J-3 displays the force and displacement results from one of the experimental trials. These natural rubber bands measured 7 inches in length with a thickness of 5/8 inch.

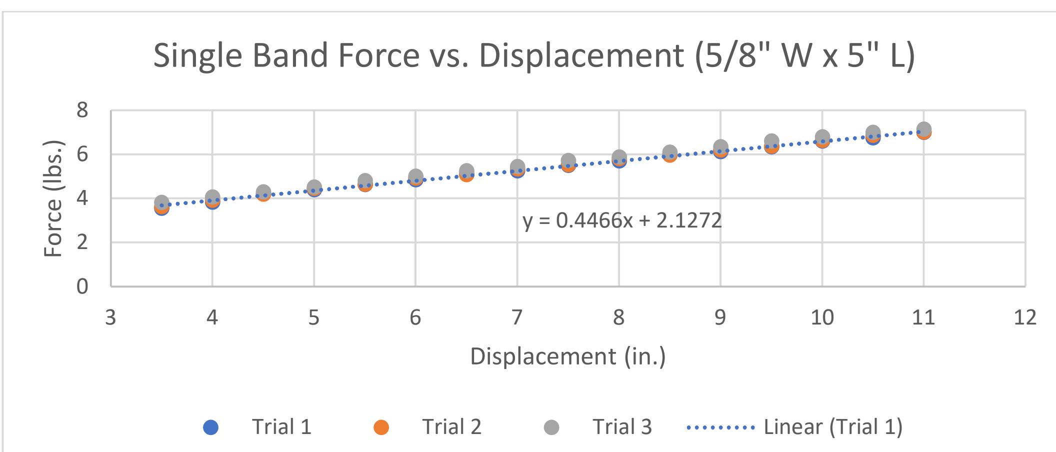

Figure J-44 The Force versus Displacement for a Natural Rubber band (5” L x 5/8” W)

Figure J-4 displays the force and displacement results from one of the experimental trials. These natural rubber bands measured 5 inches in length with a thickness of 5/8 inch. As testing progressed, so did the method of rubber band testing. By using a table clamp, one end of the rubber band was secured to a lever, and the other end was secured to the luggage scale. Using the same method as before, the bands were stretched by increments of 0.5 inches and the force was documented at each point.

As can be seen from the results above, the bands progressed in a linear manner, however, the force they yielded was not enough to meet the requirements previously set. The team conducted material research in search of alternatives that could provide a higher ‘k’ value to increase the force provided from the band.

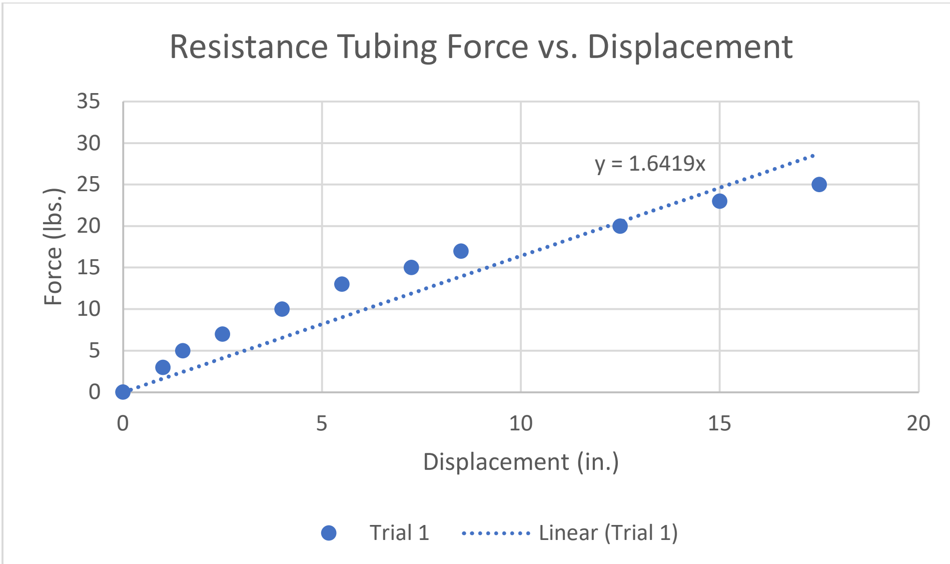

Figure J-55 The Force versus Displacement for Resistance Tubing

Figure J-5 displays the force and displacement results from one of the experimental trials. The resistance tubing measured 48 inches in length and the tubing has a diameter of ½ inch. The results for the resistance bands were favorable and showed the most potential in terms of achieving our target altitude. Given the length and strength and provided by the resistance, the bands were also tested at $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ of the overall length. Given the maximum height requirement, the resistance band would need to be cut beforehand to properly fit inside the launch mechanism raceways.

Based off these results, the team was able to determine that newly manufactured bands would be required for the mechanism, ensuring the synchronization of power-level and release-time for each foot pedal. The experiment also indicated that the rubber bands should be regularly checked for fatigue and degradation over time to ensure proper performance of the launching mechanism. Furthermore, if any quadrant’s rubber bands require replacement, all rubber bands should be replaced simultaneously to ensure proper launch mechanism performance.

Appendix K: Launch Mechanism Development Model Details

K.1: Full Assembly Details

We considered many alternative architectures, but ultimately settled on a proof-of-concept for the mechanical actuation of storing and releasing potential energy in a simple, direct way, focusing on easy setup for one operator (Figure K-1). By using four individually cocked foot pedal sliders (in blue), the total load is distributed across each and are released simultaneously against the launch platform by the synchronized release lever (both in green). For more details on the concept of operations for the cock, lock, and release mechanism see Appendix X.2. The load of the elastics is held axially between each slider against the bottom of the main launch tube housing (in grey) and the top cap (in white), with washers and eyebolts used to distribute loads across the surface of these 3D printed parts. The base part (also in white), interfaces with the launch tube housing where the flange inserts can be set with through screws, attaching the two pieces together while allowing for separation for servicing of internal parts like the elastic material, foot pedal sliders, and launch pad. The base has a thick flat bottom part with extrusions and through holes for fastening the assembly to the ground using stakes, or a wood frame using screws. The full-scale proof-of-concept was printed, assembled, and tested with both low-strength rubber bands from McMaster-Carr and different lengths of 50-pound elastic resistance tubing from GoFit. A wood frame was constructed to attach the base for rubber band testing, and elastic tube testing utilized ground stakes hammered into a grass field.

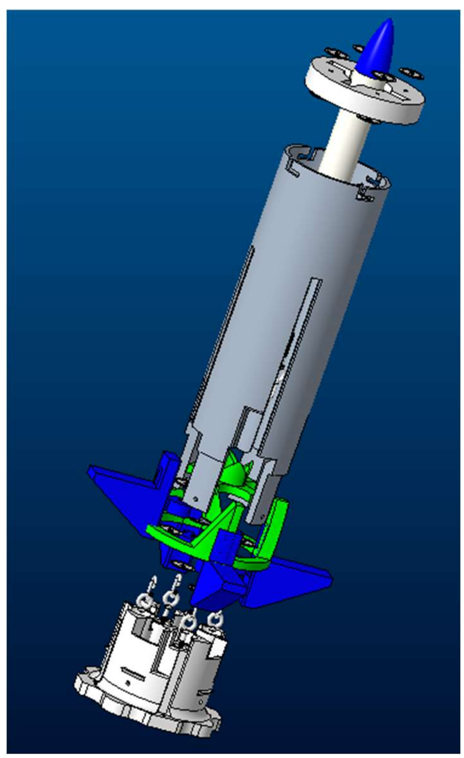

Figure K-1 Development Model Full Assembly in ProE, Exploded View.

K.2: Details on the Cock, Lock, and Release Mechanism Design and Development

The cocking and locking stages occur simultaneously for each of the foot pedals. One by one, the operator pushes down on the foot pedal until they reach the notch that corresponds to the desired launch height. Once the deflection is reached, the operator rotates the pedal slightly, sliding it into the locking notch. This notch serves to keep the pedal locked in place while the operator locks the rest of the pedals. The operator then rotates the synchronized release lever, pushing the foot pedals back into the main raceway and launching the payload to the desired height.

The multi-stage system serves multiple purposes. First, the system of individually cocking each foot pedal decreases the chances of misfires. The launch pad that the payload sits on is held up by the foot pedals. If at least one of the pedals remains at the top of the tube, the launch pad will also remain in place. It is not until the final pedal is locked into place that the tube is prepared to fire. Due to this, if a foot pedal accidentally discharges while the operator is still preparing the mechanism, the payload will not launch, and the operator will not be harmed. In addition, the multi-stage launch allows a stronger elastic band to be used. Pushing down with 100 pounds of force four times is much easier than pushing down one pedal with 400 pounds of force. The force needed to push down one of the pedals is less than the average adult's body weight, meaning that most able-bodied adults can use the mechanism alone. This allows for CLASSy to be used in many situations, and by a multitude of people, as was desired by the team. See Figure K-2 for the graphical process of cocking, locking, and releasing the mechanism.

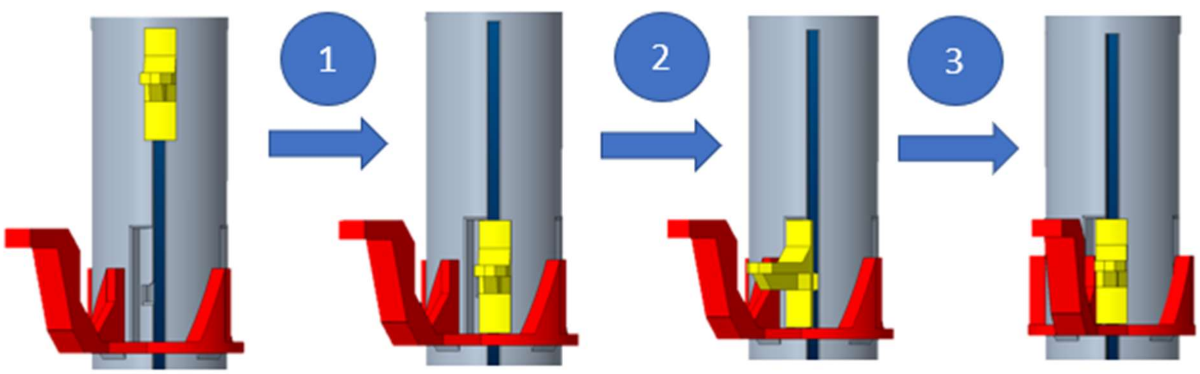

Figure K-2 The 3-stage Cock-Lock-and-Release Process.

K.3: CAD Methods

Once the launch team settled on an architecture following various brainstorming sessions and trade studies, a CAD model was designed and iterated upon. Creo Parametric (ProE) version 9.0 was used to model all parts of the development model except for basic hardware; to see how these hardware pieces interfaced with the custom parts, assembly mode in ProE was utilized to check clearances and friction fits. The circular nature of this design for many of the parts was difficult to match correct clearances, dimensioning, and tolerances. This is because 3D printing circles has a lasso-effect, where when the extruder drags the filament in a circle, it is pulled slightly inward, making any 3D printed circle approximately 2% smaller in diameter than the CAD model dimension. It was important to keep two things in mind while designing these pieces. One, how would they all be assembled and in what order; for instance, the foot pedal sliders had to be inserted from the bottom of the launch tube housing before connecting to the top plate. Two, how can these parts be modeled for ease of 3D-printing; the main launch tube housing and bottom base interface were originally one part, but then separated at the notch so support material would not be so difficult to remove and the parts easier to print.

K.4 Impact Testing of Foot Pedal Sliders

The kickback during release of the foot pedal sliders at the top of their respective raceways was enough to break through the PLA infill. A ¼-20, 2 and 3/8 inch bolt was implemented into the foot pedal slider joint through the chamfer beneath the foot pedal through the joint that connects to the circular slider (see Figure K-2). The infill of the print was increased from 35% to 50% and the number of perimeters was increased from 2 to 5 to blend fully printed perimeters together within the joint. Impact testing was conducted by dropping 6 pounds from one foot directly onto the joint, simulating the estimated impact force of a single foot pedal slider after release, which was calculated to be approximately 8 Joules of energy. The 50% infill with bolt did not fail after 3 impact tests and was then used with the rubber band testing of the development prototype. When the development prototype was tested with the elastic resistance tubing, 2 of the 50% infill foot pedal sliders with bolt failed. The infill was increased to 75% with 6 perimeters, then surviving impact testing and proved to be sufficient for the development model. It is recommended that the 3D printed foot pedal sliders be replaced with hardware-based architecture.

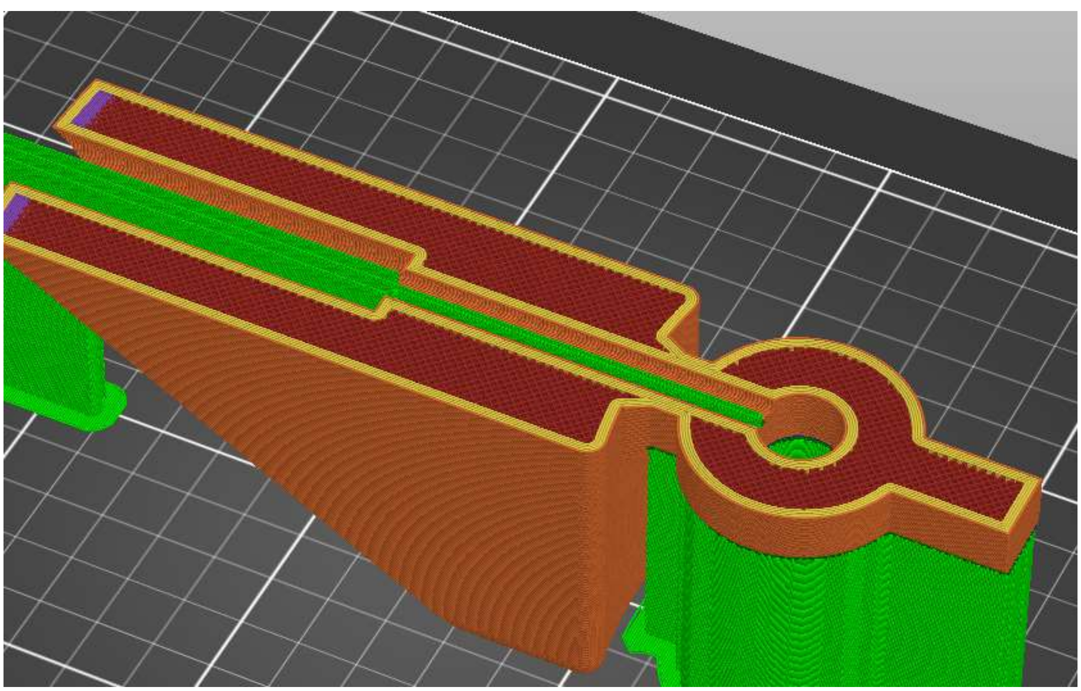

Figure K-3 Cross section of foot pedal slider part with inner-extrusion for bolt viewed in Prusa Slicer

K.5 3D Printing of Prototype Parts and Hardware Integration

3D printing of the development model parts was done on Prusa MK3, MK3S, and MK3S+ printers and the full-scale launch tube housing part as well as the synchronized release lever cam collar were printed on a Gigabot XI 3+ printer using PLA. Many small-scale models of the full assembly were printed from 30-40% scale (depending on what could fit on the 6”x6” Prusa printer beds) to ensure fits matched before printing full scale. Clearances were checked with multiple iterations, and new assembly issues were discovered while printing at small scale. This allowed the team to rapid prototype the launch mechanism development model and not waste time and PLA material on large prints before the part files were ready for full scale printing.

Hardware such as bolts within the foot pedal sliders joints, washers to distribute elastic loads across the surfaces of the interfacing custom PLA parts, and eyebolts and carabiners to directly interface with the elastic material connection points was utilized. These hardware parts were relatively inexpensive and acquired from McMaster-Carr.

K.6 Development Model CAD Screenshots and Images of Assembly

Figure K-41 The first prototype developed to prove the cock-lock-and-release mechanism.

Figure K-52 Exploded view of CLASSy Launch Mechanism Prototype

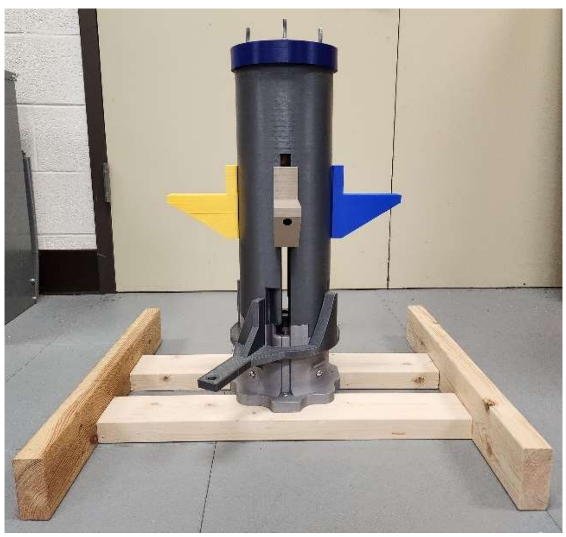

Figure K-6 Fully assembled CLASSy Launch Mechanism Prototype.

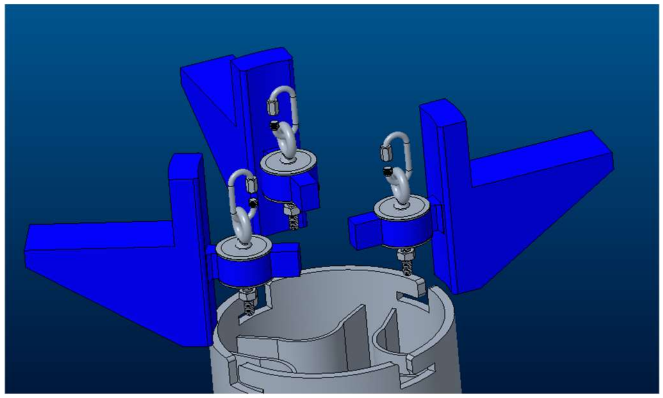

Figure K-7 CAD assembly of the launch mechanism foot pedals with hardware.

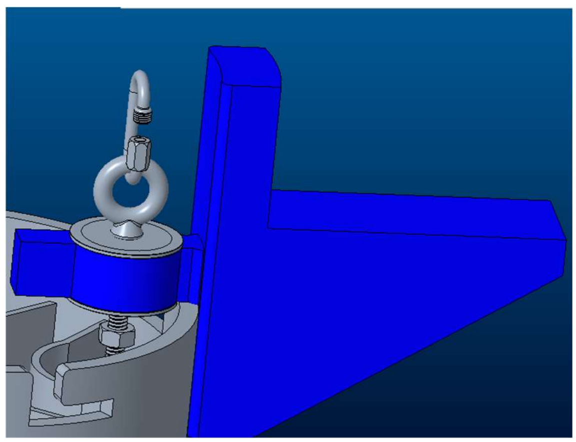

Figure K-8 3 Up-close of CAD foot pedals with hardware.

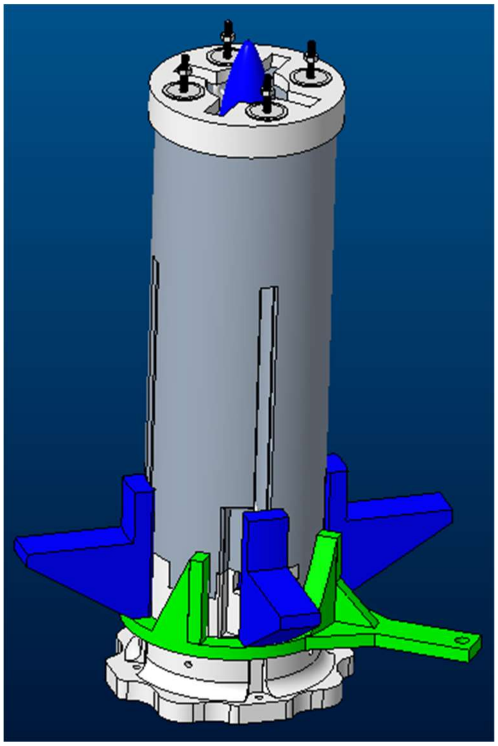

Figure K-9 Fully assembled launch mechanism.

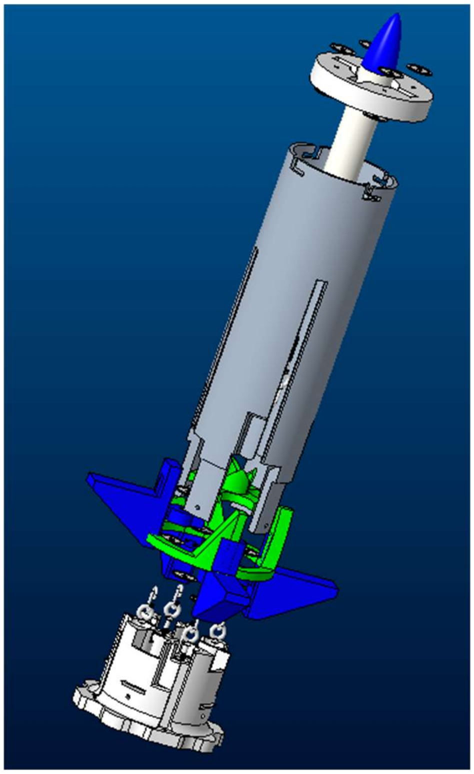

Figure K-10 Exploded view of fully assembled launch mechanism.

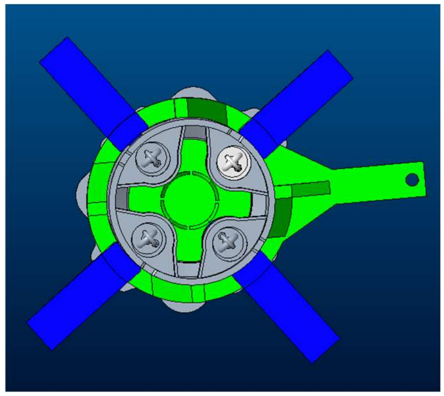

Figure K-11 Top view of launch mechanism without the top cap.

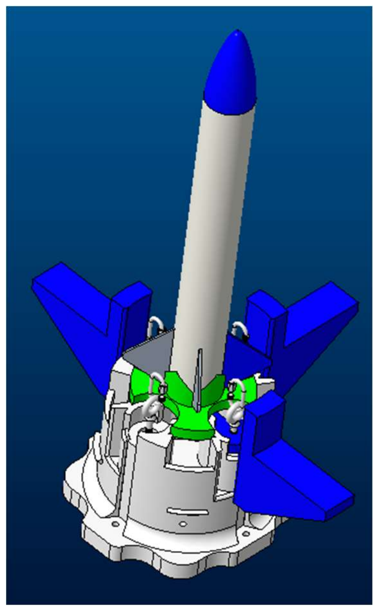

Figure K-12 Internal view of the assembled launch mechanism with the payload.

Appendix L: Launch Procedure and Pre-Launch Checklist

The following consists of Pre-Launch and Launch Procedure Checklist. This was developed by the launch team to ensure that the use of the Development Prototype was safe and easy to follow.

Pre-launch Checklist:

1. Verify that all personnel involved are wearing proper safety equipment, i.e., Safety glasses, hardhat, gloves if desired, pants, closed-toed footwear.
2. Visually verify that the individual parts of the launch mechanism have no structural deficiencies (cracks, breaks, etc...). **Disclaimer:** The parts are 3D printed PLA, so there will be seams about the circles. Do not confuse the seams with fracture.
    a. Launch tube
    b. Foot pedals
    c. Cam Collar Synchronized Release Lever (CCSRL)
    d. Base plate
    e. Launch platform
    f. Elastic band raceway
    g. Top cap
3. Visually verify that the elastic bands have no tears, fatigue, etc... and are not stretched beyond yield
4. Verify that the hardware is secure and safe
    a. Manually check if bolts are secure
    a. Visually inspect hardware for cracks or other potential hazards
5. Attach elastics to carabiners and tighten carabiner fastener, making sure the bands cannot escape the carabiner.
6. Visually verify that the elastic bands are not twisted or knotted together
7. Verify that the provided stakes are intact and have no failure points
    a. Cracks, incorrectly bent portions
8. Stake the launch mechanism into the ground firmly, checking to make sure that both the stakes and the mechanism are secured
9. Attach the launch rope to the CCSRL handle, ensuring it is secure

Launch Procedure:

1. All personnel except those physically arming the launch mechanism are 60 feet away from the launch point
2. Insert launch payload, ensuring that the fins are lined up correctly
3. Arm each pedal individually, ensuring that each pedal is locked properly prior to moving to the next pedal
4. Personnel arming the device move minimum 30 feet away from the mechanism, and/or move to the end of the launch rope
5. Once all personnel are at the proper distance(s) away, pull the launch rope, launching the payload

Appendix M: Engineering Model Details and Images

The team was unable to create a final engineering prototype, choosing rather to develop the recommendations for future development of the mechanism. The following paragraph details the interface between the pedals, sliders, and the aluminum extrusions that were mentioned in the corresponding body section of this report. The images following the description show the recommended parts that would be involved in this assembly.

First, a stack of Teflon washers (Figure M-1) with an ID equal to the width of the 80-20 aluminum framing extrusions (the new raceways, Figure M-3) are cut to fit squarely around the framing extrusion to significantly decrease friction between the sliders and raceways. Then, the ID of a tapered thrust roller bearing (Figure M-4) is bonded to the OD of the stack of Teflon washers. Bonded to the OD of the bearing is an aluminum ring clamp (Figure M-2). This clamp already has a bolt through it to tighten the part, which can be used to attach the notch key and foot pedal. We recommend that the notch key is machined out of stainless steel and the notch within the framing extrusion raceway is machined at one end to fit the key. The foot pedal may be a custom modeled 3D printed part, but cast plastic is recommended. The elastics should run on opposite sides of each raceway (to balance the torque moment between the slider and raceway to prevent binding) attached through the width of the Teflon washer stack. They could be attached by pushing through the width of the stack with a dowel to fill the elastic at one end, or by using additional hardware such as an eyebolt and carabiner to attach the elastic to the washer stack, as was used in the development prototype (see Figure K-10). This configuration allows the keyed slider to still rotate about the fixed extrusion and comply to the notched raceway and keyed slider 3-stage cock, lock and release mechanism. It has the benefit of decreasing the friction that caused energy losses in the development prototype, increased structural stability, and decreased fatigue of the system with use.

Figures M-5 and M-6 show a balsawood interior, graphite OD and fiberglass ID bonded composite cylindrical shell that is recommended to replace the main launch tube housing part. It is recommended that the final composite has a honeycomb interior, thus increasing the strength and decreasing the weight of the pack.

Figure M-12 Teflon washer
(https://www.hardin-marine.com/p-80062-1-teflon-washer.aspx)

Figure M-21 Aluminum Ring Clamp
(https://www.amazon.com/HPS-Stainless-Aluminum-Flanges-Release/dp/B08HQV8PB8)

Figure M-36: 80/20 Square Aluminum Extrusion - https://www.grainger.com/product/80-20-Framing-Extrusion-10-Series-2RCP9

Figure M-45: Tapered Thrust Roller Ball Bearing - https://www.bearingtips.com/bearings-in-motion-systems/

Figure M-54: External Tube for Engineering Prototype - Top View. Model Made of Balsa Wood, not Honeycomb.

Figure M-63: External Tube for Engineering Prototype - Front View. Model Made of Balsa Wood, not Honeycomb.